Child Immunization Tracking System (CITS) Project Documentation

The Child Immunization Tracking System (CITS) is a web-based healthcare application designed to improve the management and monitoring of child vaccination records. The system aims to replace inefficient paper-based processes with a centralized digital platform that enables healthcare workers to register children, track vaccine schedules, monitor immunization progress, and reduce missed vaccinations.

1. Project Overview

The system is designed primarily for healthcare facilities and immunization centers. It provides a simple and user-friendly environment where healthcare workers can manage child vaccination records efficiently.

The platform focuses on simplicity, reliability, scalability, and ease of use while maintaining a clean architecture suitable for future expansion.

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2. Problem Statement

Many healthcare facilities still rely heavily on manual record-keeping systems for child immunization management. These paper-based systems are often difficult to maintain, vulnerable to damage or loss, and inefficient for tracking vaccine schedules.

Common challenges include:
- Missed vaccination appointments
- Poor record management
- Difficulty monitoring immunization progress
- Lack of centralized healthcare data
- Delayed access to vaccination history

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3. Proposed Solution

The Child Immunization Tracking System will provide a centralized digital solution that:

- Registers children and stores vaccination records digitally
- Automatically assigns vaccine schedules
- Tracks completed and pending vaccinations
- Provides dashboards for monitoring progress
- Supports future integration of SMS reminders and reporting tools

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4. Project Objectives

The primary objectives of the project are:

• Reduce missed vaccinations
• Improve child immunization tracking accuracy
• Digitize healthcare record management
• Improve healthcare service delivery efficiency
• Provide a scalable system for future enhancements

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5. Target Users

The system is intended for the following users:

Administrators
Manage users, monitor reports, and oversee system activities.

Health Workers
Register children, record vaccinations, and manage schedules.

Parents/Guardians (Future Phase)
View schedules and receive vaccination reminders.

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6. Functional Requirements

The system shall:

• Allow user authentication and login
• Allow administrators to manage users
• Allow health workers to register children
• Store child demographic information
• Maintain vaccine schedules
• Record vaccination administration dates
• Display pending and completed vaccinations
• Generate dashboard statistics and summaries

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7. Non-Functional Requirements

The system should:

• Be easy to use and understand
• Provide secure authentication mechanisms
• Maintain data integrity and consistency
• Support future scalability and feature expansion
• Deliver acceptable performance and reliability
• Ensure proper access control based on user roles

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8. Technology Stack

Frontend: React.js with Tailwind CSS

Backend: Node.js with Express.js

Database: PostgreSQL

Version Control: GitHub

Testing Tools: Postman

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9. System Architecture

The system follows a three-layer architecture:

Client Layer:
Handles the user interface and user interactions through a web application.

Application Layer:
Processes requests, handles business logic, authentication, and API services.

Database Layer:
Stores user data, child records, vaccination schedules, and immunization history.

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10. Database Design

The database structure consists of the following core tables:

- Users
- Children
- Vaccines
- Vaccine Schedule
- Child Vaccinations

The database design supports structured relationships and future scalability.

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11. System Workflow

1. Healthcare worker logs into the system
2. Child information is registered
3. Vaccine schedules are assigned automatically
4. Dashboard displays upcoming vaccinations
5. Vaccinations are recorded after administration
6. The system updates immunization progress records

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12. Security Considerations

The system will implement:

- Password hashing
- Role-based access control
- Secure authentication using JWT
- Protected API routes
- Database relationship integrity

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13. Evaluation Metrics

Project performance can be measured using:

- Immunization completion rate
- Reduction in missed vaccinations
- Vaccine schedule tracking accuracy
- Notification delivery success rate
- User adoption and usability

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14. Future Enhancements

Planned future improvements include:

- SMS reminder integration
- Parent/guardian portal
- Mobile application support
- Offline synchronization support
- National healthcare integration

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15. Summary of Core Requirements

Category	Requirement
Authentication	Secure login for administrators and health workers
Child Management	Register and manage child records
Vaccination Tracking	Track vaccine administration and schedules
Database	Store records using PostgreSQL
Frontend	React.js responsive interface
Backend	Node.js REST API services
Security	JWT authentication and role-based access

Reporting	Dashboard and progress summaries

16. Current Project Status

Project Component	Status
Project Planning	Completed
System Architecture	Defined
Database Design	Completed
Backend Development	Pending
Frontend Development	Pending
Testing & Deployment	Pending

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